KAUSHIK RAJU FUNDE

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PROFESSIONAL SUMMARY

Second-year Electronics Engineering student skilled in embedded systems, ESP32/Arduino programming, and PCB design (KiCad). Experienced in circuit design, sensor interfacing, and hardware debugging through self-driven prototyping projects.

TECHNICAL SKILLS

Programming Languages: C, C++, Python, Embedded C
Microcontrollers & Interfaces: ESP32, Arduino, GPIO, PWM, ADC, UART, I2C
PCB & Circuit Design: KiCad, Schematic Capture, PCB Layout, Component Placement, Routing, Design Rule Check (DRC), 2-Layer PCB Design, Power Electronics (MOSFETs, Diodes, Transistors)
Tools: Arduino IDE, VS Code, Git/GitHub
Hardware Skills: Sensor Interfacing, OLED Display Integration, Circuit Prototyping, Prototype Testing, Hardware Debugging, Serial Debugging

PROJECTS

ESP32-Based Multi-Sensor Health Monitoring Prototype

- Designed and programmed an ESP32-based sensor prototype in Embedded C to acquire real-time data from multiple sensors over I2C
- Interfaced an OLED display for live data visualization and implemented multi-sensor communication logic
- Debugged hardware and firmware issues through serial monitoring and iterative testing
- Documented system architecture, sensor interfacing logic, and testing procedures

Automatic DC Motor Speed Controller with Protection

- Built an Arduino-based DC motor speed controller in Embedded C, driving motor speed via PWM signals from a potentiometer input
- Designed protection circuitry using a MOSFET switching stage and flyback diode to suppress inductive voltage spikes
- Tested the circuit across varying load conditions to verify stable motor control and protection behavior

PCB Design Project — Battery Level Tester

- Designed a complete battery level tester circuit in KiCad, from schematic capture through component selection and net definition
- Completed full PCB layout including component placement, routing, and Design Rule Check (DRC) for a 2-layer board
- Self-taught the complete PCB design workflow (schematic to layout) through independent study

EDUCATION

G.H. Raisoni College of Engineering, Nagpur
Bachelor of Technology (B.Tech), Electronics Engineering
Expected Graduation: 2029 | CGPA: 8.08